

ANSWERS

1.

- (a) (on melting,) bonds between molecules are broken/weakened
or molecules further apart/are able to slide over one another
kinetic energy unchanged so no temperature change
potential energy increased/changed so energy required
- (b) thermal energy/heat required to convert unit mass of solid to liquid
with no change in temperature/ at its normal boiling point
- (c) (i) thermal energy lost by water = $0.16 \times 4.2 \times 100$
= 67.2 kJ
 $67.2 = 0.205 \times L$
 $L = 328 \text{ kJ kg}^{-1}$
- (ii) more energy (than calculated) melts ice
so, (calculated) L is lower than the accepted value

2.

- (a) (Thermal) energy / heat required to convert unit mass of solid to liquid
at its normal melting point / without any change in temperature
(reference to 1 kg or to ice $\rightarrow$ water scores max 1 mark)
- (b) (i) To make allowance for heat gains from the atmosphere
- (ii) e.g. constant rate of production of droplets from funnel
constant mass of water collected per minute in beaker
(any sensible suggestion, 1 mark)
- (iii) mass melted by heater in 5 minutes = $64.7 - \frac{1}{2} \times 16.6 = 56.4 \text{ g}$
 $56.4 \times 10^{-3} \times L = 18$
 $L = 320 \text{ kJ kg}^{-1}$
(Use of $m = 64.7$, giving $L = 278 \text{ kJ kg}^{-1}$, scores max 1 mark
use of $m = 48.1$, giving $L = 374 \text{ kJ kg}^{-1}$, scores max 2 marks)

3.

- (a) increasing separation of molecules / breaking bonds between molecules B1
 (allow atoms/molecules, overcome forces)
 doing work against atmosphere (during expansion) B1
- (b) (i) 1 either bubbles produced at a constant rate / mass evaporates/lost at
constant rate
 or find mass loss more than once and this rate should be constant B1
 or temperature of liquid remains constant B1
 2 to allow/cancel out/eliminate/compensate for heat losses (to atmosphere) B1
 (do not allow 'prevent'/'stop')
- (ii) use of $\text{power} \times \text{time} = \text{mass} \times \text{specific latent heat}$ C1
 $(70 - 50) \times 5 \times 60 = (13.6 - 6.5) \times L$ C1
 $L = 845 \text{ J g}^{-1}$ A1

4.

- (a) e.g. two objects of different masses at same temperature (M1)
 same material would have different amount of heat (A1)
 e.g. temperature shows direction of heat transfer (M1)
 from high to low regardless of objects (A1)
 e.g. when substance melts/boils (M1)
 heat input but no temperature change (A1)
 any two, M1 + A1 each, max 4

- (b) (i) energy losses (to the surroundings)M1
 either increase as the temperature rises
 or rise is zero when heat loss = heat input A1
- (ii) idea of input power = maximum rate of heat loss C1
 $\text{power} = m \times c \times \Delta\theta / \Delta t$
 $54 = 0.96 \times c \times 3.7 / 60$ C1
 $c = 910 \text{ J kg}^{-1} \text{ K}^{-1}$ A1

5.

- (a) (i) 1 deg C corresponds to $(3840 - 190) / 100 \Omega$ C1
 for resistance 2300Ω , temperature is $100 \times (2300 - 3840) / (190 - 3840)$
 temperature is 42°C A1
- (ii) either $286 \text{ K} \equiv 13^\circ\text{C}$ or $42^\circ\text{C} \equiv 315 \text{ K}$ B1
 thermodynamic scale does not depend on the property of a substance M1
 so change in resistance (of thermistor) with temperature is non-linear A1

9.

- (a) (numerically equal to) quantity of (thermal) energy/heat to change state/phase of unit mass M1
at constant temperature A1
(allow 1/2 for definition restricted to fusion or vaporisation)
- (b) (i) at 70 W, mass $\text{s}^{-1} = 0.26 \text{ g s}^{-1}$ A1
at 110 W, mass $\text{s}^{-1} = 0.38 \text{ g s}^{-1}$ A1
- (ii) 1. $P + h = mL$ or substitution of one set of values C1
(110 – 70) = (0.38 – 0.26)L C1
 $L = 330 \text{ J g}^{-1}$ A1
2. either $70 + h = 0.26 \times 330$
or $110 + h = 0.38 \times 330$ C1
 $h = 17 / 16 / 15 \text{ W}$ A1

10.

- (a) (numerically equal to) quantity of heat/(thermal) energy to change state/phase of unit mass M1
at constant temperature A1
(allow 1/2 for definition restricted to fusion or vaporisation)
- (b) (i) constant gradient/straight line (allow linear/constant slope) B1
- (ii) $Pt = mL$ or power = gradient $\times L$ C1
use of gradient of graph
(or two points separated by at least 3.5 minutes) M1
 $110 \times 60 = L \times (372 - 325) \times 10^{-3} / 7.0$
 $L = 9.80 \times 10^5 \text{ J kg}^{-1}$ (accept 2 s.f.) (allow 9.8 to 9.9 rounded to 2 s.f.) A1
- (iii) some energy/heat is lost to the surroundings or vapour condenses on sides M1
so value is an overestimate A1

11.

- (a) the (thermal) energy per unit mass to raise the temperature M1
of a substance by one degree A1
(If ratio not clear for M1 mark, allow 1/2 marks for an otherwise correct answer)
- (b) (i) to allow for / determine / cancel heat transfer to / from tube / surroundings B1
(do not allow 'to stop / prevent' heat loss)

- (ii) either $P = mc\Delta\theta \pm h$
or $44.9 = 1.58 \times 10^{-3} \times c \times (25.5 - 19.5) \pm h$
or $33.3 = 1.11 \times 10^{-3} \times c \times (25.5 - 19.5) \pm h$ B1
 $(44.9 - 33.3) = (1.58 - 1.11) \times 10^{-3} \times c \times (25.5 - 19.5)$ C1
 $c = 4100 \text{ (4110) J kg}^{-1} \text{ K}^{-1}$ A1

(allow 1/3 for use of only 33.3 W, 1.11 g s⁻¹ leading to 5000 J kg⁻¹ K⁻¹)
(allow 1/3 for use of only 44.9 W, 1.58 g s⁻¹ leading to 4740 J kg⁻¹ K⁻¹)

- (c) $V_0 = 27$ or $V_{\text{rms}} = 19.1$ C1
 $33.3 = 27^2 / 2R$ or $33.3 = 19.1^2 / R$ C1
 $R = 11 \Omega$ A1

12.

- (a) no net energy transfer between the bodies
or
bodies are at the same temperature B1

- (b) (i) thermocouple, platinum/metal resistance thermometer, pyrometer B1

- (ii) thermistor, thermocouple B1

- (c) (i) change = 11.5 K B1

- (ii) final temperature = 311.2 K B1

13.

1(a)(i)	direction or rate of transfer of (thermal) energy or (if different,) not in thermal equilibrium/energy is transferred	B1
1(a)(ii)	uses a property (of a substance) that changes with temperature	B1
1(b)	- temperature scale assumes linear change of property with temperature - physical properties may not vary linearly with temperature - agrees only at fixed points Any 2 points.	B2
1(c)(i)	$Pt = mc(\Delta)\theta$	C1
	$95 \times 6 \times 60 = 0.670 \times 910 \times \Delta\theta$	M1
	$\Delta\theta = 56^\circ\text{C}$ so final temperature = $56 + 24 = 80^\circ\text{C}$	A1
	or	
	$95 \times 6 \times 60 = 0.67 \times 910 \times (\theta - 24)$	(M1)
	so final temperature or $\theta = 80^\circ\text{C}$	(A1)

1(c)(ii)	1. sketch: straight line from (0,24) to (6,80)	B1
	2. temperature drop due to energy loss = $(80 - 64) = 16^{\circ}\text{C}$	C1
	energy loss = $0.670 \times 910 \times (80 - 64) = 9800 \text{ J}$	A1
	or	
	energy to raise temperature to $64^{\circ}\text{C} = 0.670 \times 910 \times (64 - 24)$	(C1)
	$= 24400 \text{ J}$ loss = $(95 \times 6 \times 60) - 24400 = 9800 \text{ J}$	(A1)

14.

2(a)	(thermal) energy <u>per</u> (unit) mass (to cause change of state)	B1
	(energy required to cause/released in) change of state at constant temperature	B1
2(b)(i)	1. (work done on/against) the <u>atmosphere</u>	B1
	2. <u>water</u> as it turns from liquid to vapour	M1
	as potential energy of <u>molecules</u> increases	A1
	or	
	<u>surroundings</u> as its temperature rises	(M1)
	as energy is lost/transferred to surroundings	(A1)
2(b)(ii)	$VI - h = M/t \times L$ (where h = power loss) or $L = (VI t - Q)/M$ (where Q = energy loss)	C1
	$(14.2 \times 6.4) - (11.5 \times 5.2) = (9.1 - 5.0) \times L/300$ or $L = [(14.2 \times 6.4) - (11.5 \times 5.2)] \times 300 / (9.1 - 5.0)$	C1
	$L = 2300 \text{ J g}^{-1}$	A1

15.

3(a)	(during melting,) bonds between atoms/molecules are broken	B1
	potential energy of atoms/molecules is increased	B1
	no/little work done so required input of energy is thermal	B1
3(b)(i)	$(\Delta Q =) mc\Delta\theta$	C1
	loss = $(160 \times 0.910 \times 15) + (330 \times 4.18 \times 15)$ $= 2.3 \times 10^4 \text{ J}$	A1
3(b)(ii)	$2.3 \times 10^4 = (48 \times 2.10 \times 18) + 48L + (48 \times 4.18 \times 23)$	C1
	$48L = 1.66 \times 10^4$	A1
	$L = 350 \text{ J g}^{-1}$	

16.

3(a)	(thermal) energy per unit mass (to cause change of state)	B1
	(energy transfer during) change of state between solid and liquid at constant temperature	B1
3(b)(i)	Any one from: - rate of increase in mass (of beaker and water) is constant - level of water rises at a constant rate - volume of water (in beaker) increases at a constant rate - constant time between drops - constant rate of dripping	B1
3(b)(ii)	(electrical power supplied $\Rightarrow$) 12.8×4.60 ($= 58.9 \text{ W}$)	C1
	(rate of transfer to ice $\Rightarrow$) $[(185.0 - 121.5) \times 332] / [5.00 \times 60]$ ($= 70.3 \text{ W}$)	C1
	1. rate $= 70.3 \text{ W}$	A1
	2. rate $= 70.3 - 58.9$ $= 11.4 \text{ W}$	A1

17.

3(a)	(thermal) energy per (unit) mass (to change state)	B1
	(heat transfer during) change of state at constant temperature	B1
3(b)(i)	32 g	A1
3(b)(ii)	temperature difference (between liquid and surroundings) does not change	B1
3(b)(iii)	$VIt = mL$	C1
	$230 \times 1.2 \times 60 \times 10 = (56 \times L) + H$ or $190 \times 1.0 \times 60 \times 10 = (32 \times L) + H$	C1
	$86 \times 600 = (56 - 32) \times L$	C1
	or	
	$230 \times 1.2 = (56 \times L) / (60 \times 10) + P$ or $190 \times 1.0 = (32 \times L) / (60 \times 10) + P$	(C1)
	$276 - 190 = (24 \times L) / 600$	(C1)
	$L = 2200 \text{ J g}^{-1}$	A1
3(b)(iv)	$230 \times 1.2 \times 600 = (56 \times 2150) + H$ or $190 \times 1.0 \times 600 = (32 \times 2150) + H$	C1
	$H = 45200$ rate $= 45200 / 600$ $= 75 \text{ W}$	A1
	or	
	$230 \times 1.2 = (56 \times 2150) / (60 \times 10) + P$ or $190 \times 1.0 = (32 \times 2150) / (60 \times 10) + P$	(C1)
	rate ($= P$) $= 75 \text{ W}$	(A1)

18.

3(a)	(thermal) energy per unit mass (to change state)	B1
	change of state without any change of temperature	B1
3(b)(i)	140 g	A1
3(b)(ii)	temperature difference (between apparatus and surroundings) does not change	B1
3(b)(iii)	$VIt = mL$	C1
	$((15.1 \times 3.6) + R) \times 600 = 140 \times L$ or $((7.3 \times 1.8) + R) \times 600 = 65 \times L$	C1
	$41.22 \times 600 = 75 \times L$	C1
	$L = 330 \text{ J g}^{-1}$	A1
3(b)(iv)	$15.1 \times 3.6 \times 600 = (140 \times 330) - H$ or $7.3 \times 1.8 \times 600 = (65 \times 330) - H$	C1
	$H = 13600$	A1
	rate of gain = $13600 / 600$ $= 23 \text{ W}$	

19.

3(a)	Any 2 from: - particles / atoms / molecules / ions (very) close together / touching - regular, repeating pattern - vibrate about a fixed point	B2
3(b)	(much) greater <u>increase</u> in spacing of molecules (for vaporisation compared with fusion)	B1
3(c)(i)	-100°C	B1
3(c)(ii)	time = $8.5 - 3.0$ $= 5.5 \text{ min}$	C1
	$Pt = mL$ energy = power $\times$ time = $150 \times 5.5 \times 60$ $= 49\,500 \text{ J}$ $L = \frac{E}{m}$ $= \frac{49500}{0.045}$	C1
	$= 1100 \text{ kJ kg}^{-1}$	A1
3(c)(iii)	gas has a higher specific heat capacity (than liquid)	B1

20.

2(a)	- resistance of a metal - volume of a gas at constant pressure - e.m.f. of a thermocouple <i>Any two points, 1 mark each</i>	B2
2(b)(i)	$Q = mc\Delta T$	C1
	evidence of realisation that Q lost by water = Q gained by mercury	C1
	$18.7 \times 4.18 \times (37.4 - T) = 6.94 \times 0.140 \times (T - 23.0)$	C1
	$T = 37.2\text{ }^{\circ}\text{C}$	A1
2(b)(ii)	use a liquid with a lower (specific) heat capacity (than mercury) or use a smaller mass of mercury	B1
2(c)(i)	depends on properties of a real substance	B1
	$0\text{ }^{\circ}\text{C}$ is not absolute zero	B1
2(c)(ii)	ideal gas	B1

21.

3(a)	no <u>net</u> thermal energy is transferred (between them)	B1
3(b)(i)	variation (of density with temperature) is linear or each temperature has a unique value of density	B1
3(b)(ii)	- variation (of density with temperature) is not linear - region where the density does not vary with temperature - different temperatures have the same density <i>Any two points, 1 mark each</i>	B2
3(c)(i)	boiling point = $80\text{ }^{\circ}\text{C}$	A1
3(c)(ii)	$Q = Pt$ and $t = 21\text{ s}$ (thermal energy supplied = $810 \times 21 = 17000\text{ J}$)	C1
	$c = Q / m\Delta\theta$	C1
	thermal energy absorbed by beaker = $42 \times 0.84 \times (80 - 25)$ (= 1940 J)	C1
	s.h.c. of liquid = $[(810 \times 21) - (42 \times 0.84 \times (80 - 25))] / [120 \times (80 - 25)]$ $= 2.3\text{ J g}^{-1}\text{ K}^{-1}$	A1
3(d)	sketch: straight diagonal line from $25\text{ }^{\circ}\text{C}$ to $100\text{ }^{\circ}\text{C}$ and then horizontal at $100\text{ }^{\circ}\text{C}$	B1
	straight diagonal line starting at $25\text{ }^{\circ}\text{C}$ with gradient approximately half that of the original line	B1